

Thermodynamics and statistical mechanics

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1 Thermodynamics

1.1 What is Thermodynamics?

Thermodynamics offers a framework for describing macroscopic systems in thermal equilibrium, providing a bridge between microscopic physical phenomena and observable macroscopic behavior. Rooted in classical mechanics, thermodynamics extends its reach through statistical mechanics, which connects the behavior of individual particles to bulk properties. The essence of thermodynamics lies in its use of state functions or thermodynamic coordinates, such as temperature, pressure, and volume, that encapsulate the equilibrium state of the system. Once a system reaches this equilibrium, it remains stable over time, with no further net changes in its thermodynamic variables. Through this approach, thermodynamics serves as a powerful tool for simplifying and predicting the behavior of complex systems.

1.2 The zeroth law

Theorem 1 *If two systems, A and B, are each in equilibrium with a third system, C, then A and B are also in equilibrium with each other.*

The zeroth law establishes a transitive relationship among thermodynamic systems and defines the empirical temperature Θ , as a state function. Specifically, systems that are in thermal equilibrium with each other must share the same temperature. To formalize this, consider the thermodynamic coordinates of systems A, B, and C as $\{A_i\} = \{A_1, A_2, A_3, \dots\}$, $\{B_i\} = \{B_1, B_2, B_3, \dots\}$, and $\{C_i\} = \{C_1, C_2, C_3, \dots\}$, respectively. If system A is in equilibrium with system C, we can express this constraint as:

$$f_{AC}(\{A_i\}, \{C_i\}) = 0 \quad (1)$$

Similarly, if system B is in equilibrium with system C, we have the condition:

$$f_{BC}(\{B_i\}, \{C_i\}) = 0 \quad (2)$$

According to the zeroth law, systems A and B must also be in equilibrium, leading to the following relation:

$$f_{AB}(\{A_i\}, \{B_i\}) = 0 \quad (3)$$

Now, let us consider a particular variable, C_j of system C. We can express C_j in terms of the other variables of A and C as:

$$C_j = F_{AC}(\{A_i\}, \{C_i\} \setminus C_j) \quad (4)$$

Similarly, for system B:

$$C_j = F_{BC}(\{B_i\}, \{C_i\} \setminus C_j) \quad (5)$$

And for the direct relationship between A and B:

$$C_j = F_{AB}(\{A_i\}, \{B_i\}) \quad (6)$$

By comparing these equations, we observe that the remaining variables $\{C_i\} \setminus C_j$ can be eliminated, demonstrating that:

$$\Theta(\{A_i\}) = \Theta(\{B_i\}) = \Theta \quad (7)$$

Thus, systems A and B share the same temperature, specifying the equation of state.

1.3 The first law

Theorem 2 *conservation of energy gives $\vec{dq} = dE - \vec{dw}$, where $dE = \sum_i \partial_i E dX_i$ can be obtained by differentiation, while $\vec{dq}$ and $\vec{dw}$ generally cannot.*

A quasi-static process ensures that the transformation occurs slowly enough to maintain the system in equilibrium. The state functions $\{\vec{X}\}$ can be categorized into generalized displacements $\{\vec{x}\}$ and their corresponding generalized forces $\{\vec{J}\}$, enabling an infinitesimal quasi-static transformation.

$$dW = \sum_i J_i dx_i \quad (8)$$

Displacements are extensive quantities, proportional to the size of the system, while forces, their conjugate variables, are intensive and independent of the size of the system. In particular, intensive quantities serve as indicators of equilibrium. Table 1 lists common examples of such conjugate pairs. Response functions characterize the macroscopic behaviour of a system and are experimentally measured through the variation of thermodynamic coordinates under external probes. For example, heat capacities are determined by the change in temperature upon addition of heat to the system.

$$C_V = \left(\frac{dq}{dT} \right)_V = \left(\frac{dE - dW}{dT} \right)_V = \left(\frac{dE + pdV}{dT} \right)_V = \left(\frac{\partial E}{\partial T} \right)_V \quad (9)$$

$$C_P = \left(\frac{dq}{dT} \right)_P = \left(\frac{dE - dW}{dT} \right)_P = \left(\frac{dE + pdV}{dT} \right)_P = \left(\frac{\partial E}{\partial T} \right)_P + P \left(\frac{\partial V}{\partial T} \right)_P \quad (10)$$

Force constants, such as the isothermal compressibility of a gas $\kappa_T = -\frac{1}{V} \left(\frac{\partial V}{\partial p} \right)_T$, and the magnetic susceptibility $\chi_T = \frac{1}{V} \left(\frac{\partial M}{\partial B} \right)_T$, quantify the system's response by measuring the ratio of displacement to forces.

Thermal response functions examine changes in thermodynamic coordinates with respect to temperature. For instance, the thermal expansivity of a gas is given by $\alpha_p = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_p$.

System	Force	Intensive	Displacement	Extensive
Wire	tension	F	length	L
Film	surface tension	$\mathcal{S}$	area	A
Fluid	pressure	$-p$	volume	V
Magnet	magnetic field	H	magnetization	M
Dielectric	electric field	E	polarization	P
Chemical reaction	chemical potential	μ	particle number	N
Heat	temperature	T	entropy	S

Table 1: Intensive quantities and their extensive conjugates in different systems

1.4 The second law

Theorem 3 *In the isolated system, the total entropy changes cannot be negative $\Delta S_t \geq 0$.*

Corollary 3.1 Kelvin's statement. *No process is possible whose sole result is the complete conversion of heat into work.*

Corollary 3.2 Clausius's statement. *No process is possible whose sole result is the transfer of heat from a colder to a hotter body.*

We analyze a system connected to a Carnot engine, which functions by extracting heat from a high-temperature reservoir and discharging it into a low-temperature reservoir. Denote the ambient temperature as T_0 , while the system temperature is T . The heat transfer in this scenario is governed by the equation: $\dot{d}q_R = T_0 \frac{\dot{d}q}{T}$, where $\dot{d}q_R$ represents the heat extracted by the Carnot engine and $\dot{d}q$ denotes the heat absorbed by the system.

According to the first law of thermodynamics, the total work W performed by both the Carnot engine and the system is equal to the total heat input, $Q_R = \oint \dot{d}q_R$. Nevertheless, Kelvin's statement of the second law dictates that achieving a positive work output over a complete cycle is impossible, as some heat must continually be expelled. Consequently, the total work must adhere to $W = Q_R = \oint \dot{d}q_R = \oint T_0 \frac{\dot{d}q}{T} \leq 0$. We derive the inequality: $\oint \frac{\dot{d}q}{T} \leq 0$, which validates the proof, aligning with the second law of thermodynamics and the Clausius inequality.

Theorem 4 Clausius's theorem. *For any cyclic transformation (reversible or not), $\oint \frac{\dot{d}q}{T} \leq 0$, where $\dot{d}q$ is the heat increment supplied to the system at temperature T .*

Clausius's theorem offers fundamental insights into the existence of the state function known as entropy. To investigate this, consider formulating an idealized thermodynamic cycle in which the transformation from state A to state B is arbitrary, but the return path from B to A is reversible. This setup yields the inequality: $\oint_A^B \frac{\dot{d}q}{T} + \oint_B^A \frac{\dot{d}q_{\text{rev}}}{T} \leq 0$. The existence of entropy S as a state function can be inferred because this integral is path independent and depends solely on the end points. We define the change in entropy as:

$$S(B) - S(A) \equiv \int_A^B \frac{\dot{d}q_{\text{rev}}}{T} \quad (11)$$

Therefore, for any process connecting A and B , the inequality $\int_A^B \frac{\dot{d}q}{T} \leq S(B) - S(A) = \Delta S$ holds. In differential form, this implies $dS \geq \frac{\dot{d}q}{T}$ for all processes. As the system approaches equilibrium, the net heat exchange $\dot{d}q = 0$, yielding $\delta S \geq 0$. Consequently, **for an adiabatic system, entropy reaches its maximum at equilibrium, as spontaneous internal changes solely increase entropy.**

In a reversible (or quasi-static) transition, $dq = TdS$ and $dW = \sum_i J_i dx_i$, then according to the first law of thermodynamics,

$$dE = dq + dW = TdS + \sum_i J_i dx_i \quad (12)$$

This expression establishes a precise relationship between mechanical and thermal interactions. If there are n distinct mechanisms for doing work on the system, represented by n conjugate pairs (J_i, x_i) , then $n + 1$ independent variables suffice to describe the system, with the thermal pair being S and T . The partial derivatives of entropy characterize the temperature and forces within the system:

$$\left(\frac{\partial S}{\partial E}\right)_X = \frac{1}{T} \quad \text{and} \quad \left(\frac{\partial S}{\partial x_i}\right)_{E, X_{j \neq i}} = -\frac{J_i}{T} \quad (13)$$

1.4.1 Equilibrium and thermodynamic potential

Enthalpy For any set of displacements $\vec{x}$ with a constant external force $\vec{J}$, the input to the system works satisfies $dW \leq \vec{J} \cdot \delta \vec{x}$. The equality is met in a quasi-static process, although some external work might be dissipated. In absence of thermal transfer $dQ = 0$, the first law of thermodynamics provides $\delta E \leq \vec{J} \cdot \delta \vec{x}$, thus we define enthalpy as:

$$\delta H \leq 0, \quad \text{where} \quad H = E - \vec{J} \cdot \vec{x} \quad (14)$$

This presents the concept of enthalpy, merging the system's internal energy with the work against external forces. The equilibrium changes of H derive from the Legendre transformation:

$$dH = dE - d(\vec{J} \cdot \vec{x}) = TdS - \vec{x} \cdot d\vec{J} \quad (15)$$

Here, d denotes equilibrium state changes, while δ indicates non-equilibrium variations. The coordinate set $(S, \vec{J})$ is a natural choice to describe the enthalpy. The system's generalized displacements are described by:

$$x_i = -\left(\frac{\partial H}{\partial J_i}\right)_{S, J_{i \neq j}} \quad \text{and} \quad T = \left(\frac{\partial H}{\partial S}\right)_J \quad (16)$$

The temperature dependence of the enthalpy is directly related to the heat capacity under constant external force, as shown in equation 10. Specifically, the heat capacity at constant pressure C_p is related to the enthalpy's temperature variation:

$$C_p = \left(\frac{dq}{dT}\right)_p = \left(\frac{dE + pdV}{dT}\right)_p = \left(\frac{d(E + pV)}{dT}\right)_p = \left(\frac{dH}{dT}\right)_p \quad (17)$$

Helmholtz free energy is instrumental in characterizing isothermal processes, where the temperature remains invariant and no mechanical work is done, i.e., $dW = 0$. Based on Clausius's theorem (see theorem 4), the heat absorbed by a system at constant temperature T meets the condition $dq \leq TdS$. Consequently, $\delta E = dq + dW \leq TdS$. This correlation leads, through the first law of thermodynamics, to changes in the Helmholtz free energy, thereby substantiating the definition of Helmholtz free energy F , formulated through a Legendre transformation.

$$\delta F \leq 0, \quad \text{with} \quad F = E - TS \quad (18)$$

$$dF = dE - d(TS) = -SdT + \vec{J} \cdot d\vec{x} \quad (19)$$

Accordingly, Helmholtz free energy is most effectively represented using the coordinate set $(T, \vec{x})$. From F , the equilibrium forces $\vec{J}$ and entropy S can be computed as:

$$J_i = \left(\frac{\partial F}{\partial x_i}\right)_{T, X_{i \neq j}} \quad \text{and} \quad S = -\left(\frac{\partial F}{\partial T}\right)_X \quad (20)$$

Moreover, the internal energy E can be derived from Helmholtz free energy by the expression:

$$E = F + TS = F - T \left(\frac{\partial F}{\partial T}\right)_X = -T^2 \left(\frac{\partial(F/T)}{\partial T}\right)_X \quad (21)$$

Gibbs free energy holds significant importance for isothermal processes involving mechanical work under constant external forces. The natural inequalities for the work and heat introduced into the system are given by $dW \leq \vec{J} \cdot \delta \vec{x}$ and $dQ \leq T\delta S$. Hence, the total energy variation follows $\delta E \leq T\delta S + \vec{J} \cdot \delta \vec{x}$. This forms the basis for defining the Gibbs free energy and its differential form, which inherently depends on the coordinate set $(T, \vec{J})$.

$$\delta G \leq 0, \quad \text{where} \quad G = E - TS - \vec{J} \cdot \vec{x} \quad (22)$$

$$dG = dE - d(TS) - d(\vec{J} \cdot \vec{x}) = -SdT - \vec{x} \cdot d\vec{J} \quad (23)$$

Grand potential is utilized to characterize systems at chemical equilibrium, specifically when mechanical work is not present. In these systems, changes in particle numbers alter the internal energy, expressed as chemical work $dW = \vec{\mu} \cdot d\vec{N}$. Here, $\vec{N} = \{N_1, N_2, \dots\}$ denotes the particle numbers of each species, while $\vec{\mu} = \{\mu_1, \mu_2, \dots\}$ are the pertinent chemical potentials, indicating the work necessary to add more particles to the system. The grand potential Γ is defined via the Legendre transformation.

$$\Gamma = E - TS - \vec{\mu} \cdot \vec{N} \quad (24)$$

At chemical equilibrium, the grand potential $\Gamma(T, \vec{\mu}, \vec{x})$ reaches a minimum, and its differential variation is expressed as

$$d\Gamma = -SdT + \vec{J} \cdot d\vec{x} - \vec{N} \cdot d\vec{\mu} \quad (25)$$

1.4.2 Mathematical results

Extensivity When considering chemical work, variations of the extensive coordinates of the system are generalized by extending equation 12.

$$dE = TdS + \vec{J} \cdot d\vec{x} + \vec{\mu} \cdot d\vec{N} \quad (26)$$

For constant intensive coordinates, the extensive properties scale with the size or the number of particles.

$$E(\lambda S, \lambda \vec{x}, \lambda \vec{N}) = \lambda E(S, \vec{x}, \vec{N}) \quad (27)$$

Taking the derivative of the above equation with respect to λ and evaluating at $\lambda = 1$, we obtain

$$\left(\frac{\partial E}{\partial S}\right)_{x, N} S + \sum_i \left(\frac{\partial E}{\partial x_i}\right)_{S, x_i \neq j, N} x_i + \sum_i \left(\frac{\partial E}{\partial N_i}\right)_{S, x, N_i \neq j} N_i = E(S, \vec{x}, \vec{N}) \quad (28)$$

Identifying the partial derivatives in the above equation from equation 26 as T , J_i , and μ_i , respectively, we have **Euler equation**.

$$E = TS + \vec{J} \cdot \vec{x} + \vec{\mu} \cdot \vec{N} \quad (29)$$

Combining the variations from equation 29 with those from equation 26 imposes a constraint on the variations of intensive coordinates, known as the **Gibbs-Duhem relation**.

$$SdT + \vec{x} \cdot d\vec{J} + \vec{N} \cdot d\vec{\mu} = 0 \quad (30)$$

Maxwell relations The first-order derivatives of thermodynamic potentials yield crucial physical properties, such as temperature, pressure, and other intensive variables. By taking these derivatives with respect to different pairs of variables, one derives Maxwell relations, which stem from the commutative nature of second-order mixed partial derivatives, namely $\partial_x \partial_y f(x, y) = \partial_y \partial_x f(x, y)$. For example, consider the internal energy $E(S, x)$ given by $TdS + Jdx$.

$$\frac{\partial^2 E}{\partial S \partial x} = \frac{\partial^2 E}{\partial x \partial S} = \left(\frac{\partial T}{\partial x}\right)_S = \left(\frac{\partial J}{\partial S}\right)_x \quad (31)$$

Given that $\left(\frac{\partial y}{\partial x}\right) = \left(\frac{\partial x}{\partial y}\right)^{-1}$, the prior equation can be rewritten as follows:

$$\left(\frac{\partial x}{\partial T}\right)_S = \left(\frac{\partial S}{\partial J}\right)_x \quad (32)$$

Chain rule of differentiation is a fundamental principle in thermodynamics for interrelating partial derivatives of various variables. For example, consider the entropy S , volume V , energy E , and particle number N .

$$\left(\frac{\partial S}{\partial V}\right)_{E,N} = \frac{P}{T} = -\frac{(\partial E/\partial V)_{S,N}}{(\partial E/\partial S)_{V,N}} \quad (33)$$

The above expression can be reformulated into:

$$\left(\frac{\partial S}{\partial V}\right)_{E,N} \left(\frac{\partial E}{\partial S}\right)_{V,N} \left(\frac{\partial V}{\partial E}\right)_{S,N} = -1 \quad (34)$$

This relationship exemplifies how partial derivatives of thermodynamic variables are interconnected to describe variations in state functions and their reliance on diverse thermodynamic coordinates.

1.4.3 Stability

A particle moving freely in an external potential $U(x)$ dissipates energy and eventually settles at an equilibrium corresponding to the minimum of U . Stability at this minimum is guaranteed by the condition $\frac{d^2U}{dx^2} > 0$, to ensure that the system remains in equilibrium under small perturbations. When an external force J is applied, the enthalpy $H = U - Jx$ is minimized, effectively tilting the potential. At the new equilibrium point $x_{eq}(J)$, the second derivative of the enthalpy becomes $\frac{d^2H}{dx^2} = \frac{d^2U}{dx^2} - \frac{dJ}{dx} = \frac{d^2U}{dx^2} > 0$. This implies that only the convex regions of the potential $U(x)$ are physically accessible, corresponding to the stable equilibrium.

In systems with more than one mechanical coordinate, stability requires that any small displacement δx results in an increase in the energy or enthalpy of the system. This can be written as the following condition on the joint second derivatives of U .

$$\sum_{i,j} \frac{\partial^2 U}{\partial x_i \partial x_j} \delta x_i \delta x_j > 0 \quad (35)$$

This expression guarantees that small deviations from equilibrium increase the potential energy, indicating stability. The corresponding change in force due to these displacements is given by

$$\delta J_i = \delta \left(\frac{\partial U}{\partial x_i} \right) = \sum_j \frac{\partial^2 U}{\partial x_i \partial x_j} \delta x_j \quad (36)$$

Thus, the stability condition is equivalent to

$$\sum_i \delta J_i \delta x_i > 0 \quad (37)$$

This ensures that the forces resulting from small perturbations act to restore equilibrium. In the presence of thermal and chemical forces, the stability criterion can be generalized to include temperature T , entropy S , chemical potentials μ_i , and particle numbers N_i . The general stability condition becomes

$$\delta T \delta S + \sum_i \delta J_i \delta x_i + \sum_i \delta \mu_i \delta N_i > 0 \quad (38)$$

This expression ensures that perturbations in mechanical, thermal, or chemical variables lead to an increase in the appropriate potential, maintaining system stability.

1.5 The third law

Theorem 5 *The entropy of all systems at zero absolute temperature is a universal constant that can be taken to be zero; $\lim_{T \rightarrow 0} S(\vec{x}, T) = 0$.*

2 Classical statistical mechanics

2.1 The microcanonical ensemble

We assume that in the isolated system, all microstates have the same probability. The probability of each state can be obtained by counting the number of total microstates.

$$P(\nu) = \begin{cases} \frac{1}{\Omega(E, V, N)} & \text{if } E_\nu = E, V_\nu = V, N_\nu = N \\ 0 & \text{otherwise} \end{cases} \quad (39)$$

The entropy of this uniform probability distribution is given by

$$S(N, V, E) = k_B \ln \Omega(N, V, E) \quad (40)$$

2.2 The canonical ensemble

We can extend the framework to open systems, for example, one that can exchange energy with a bath. The total system is isolated, composed of a system and a bath. The state is shown as $\{\nu_t\} = \{\nu, \nu_B\}$, with the total energy $E_t = E(\nu) + E_B(\nu_B)$. For a given ν , there are several states of bath.

$$P(\nu) = \sum_{\nu_B} P(\nu, \nu_B) = \frac{\Omega_B(E_B)}{\Omega_t} = \frac{\Omega_B(E_t - E(\nu))}{\Omega_t} \quad (41)$$

The system energy is negligible compared to the total energy of isolated system. We can do the Taylor expansion on the system energy.

$$\ln \Omega_B(E_t - E(\nu)) = \ln \Omega(E_t) - E \left(\frac{\partial \ln \Omega_B}{\partial E_B} \right)_{E=0} + \dots \quad (42)$$

where $\left(\frac{\partial \ln \Omega_B}{\partial E_B} \right)$ is the property of the bath β . As the thermodynamic change on the entropy $dS = \frac{1}{T}dE + \frac{p}{T}dV - \frac{\mu}{T}dN$, β also indicates the inverse temperature $\beta = \frac{1}{k_B T}$. Therefore, the probability of the microstate in the system is $P(\nu) \propto e^{-\beta E(\nu)}$.

The partition function of the system $Q(N, V, T) = \sum_{\nu} e^{-\beta E(\nu)}$ is the normalization constant of the probability of each microstate in canonical ensemble.

$$P(\nu) = \begin{cases} \frac{e^{-\beta E(\nu)}}{Q} & \text{if } T_\nu = T, V_\nu = V, N_\nu = N \\ 0 & \text{otherwise} \end{cases} \quad (43)$$

The probability of each energy in the canonical ensemble involves the degeneracy of the state.

$$P(\epsilon) = \sum_{\nu} P(\nu) \delta(E - \epsilon) = \frac{\Omega e^{-\beta \epsilon}}{Q} = \frac{1}{Q} \exp \left\{ \frac{S}{k_B} - \frac{\epsilon}{k_B T} \right\} = \frac{e^{-\beta F}}{Q} \quad (44)$$

As $P(\epsilon)$ is sharply peaked at the most probable energy E^* , minimizing the Helmholtz free energy F , we can calculate F using the partition function.

$$F(N, V, T) = -k_B T \ln Q(N, V, T) \quad (45)$$

The average and fluctuation of energy can be calculated from the partition function, by noting that $Q(\beta)$ is related to the moment-generating function for E .

$$-\frac{\partial Q}{\partial \beta} = \sum_{\nu} E e^{-\beta E} \quad \text{and} \quad \frac{\partial^2 Q}{\partial \beta^2} = \sum_{\nu} E^2 e^{-\beta E} \quad (46)$$

The cumulants of E are generated by $\ln Q(\beta)$

$$-\frac{\partial \ln Q}{\partial \beta} = -\frac{1}{Q} \frac{\partial Q}{\partial \beta} = \sum_{\nu} E \frac{e^{-\beta E}}{Q} = \langle E \rangle \quad (47)$$

$$\frac{\partial^2 \ln Q}{\partial \beta^2} = -\frac{1}{Q^2} \left(\frac{\partial Q}{\partial \beta} \right)^2 + \frac{1}{Q} \frac{\partial^2 Q}{\partial \beta^2} = -\langle E \rangle^2 + \langle E^2 \rangle = \langle (\delta E)^2 \rangle \quad (48)$$

$$= -\frac{\partial \langle E \rangle}{\partial \beta} = k_B T^2 \frac{\partial \langle E \rangle}{\partial T} = k_B T^2 C_v \quad (49)$$

2.3 The isothermal-isobar ensemble

The internal energy changes with the addition of both heat and work; thus thermodynamic coordinates $\vec{x}$ appear as additional random variables. We can extend the system contact with a reservoir at the same temperature with fixed pressure. The probability of microstates of the combined system can be calculated with the canonical formalism.

$$P(\nu) = \begin{cases} \frac{e^{-\beta(E(\nu)+pV)}}{Z} & \text{if } T_\nu = T, p_\nu = p, N_\nu = N \\ 0 & \text{otherwise} \end{cases} \quad (50)$$

where the Gibbs partition function of system is $Z(N, p, T) = \sum_\nu e^{-\beta(E(\nu)+pV)}$. The probability of each energy in the Gibbs canonical ensemble is calculated based on the equation 44.

$$P(\epsilon) = \frac{e^{-\beta(F+pV)}}{Z} = \frac{e^{-\beta G}}{Z} \quad (51)$$

As $P(\epsilon)$ is sharply peaked at the most probable energy F^* , minimizing the Gibbs free energy G , we can calculate G using the Gibbs partition function.

$$G(N, p, T) = -k_B T \ln Z(N, p, T) \quad (52)$$

The average and fluctuation of coordinate V can be calculated from the Gibbs partition function, similar to the equation 46.

$$-\frac{\partial Z}{\partial \beta p} = \sum_\nu V e^{-\beta(E+pV)} \quad \text{and} \quad \frac{\partial^2 Z}{\partial (\beta p)^2} = \sum_\nu V^2 e^{-\beta(E+pV)} \quad (53)$$

The cumulants of V are generated by $\ln Z(\beta p)$.

$$-\frac{\partial \ln Z}{\partial \beta p} = -\frac{1}{Z} \frac{\partial Z}{\partial \beta p} = \sum_\nu V \frac{e^{-\beta(E+pV)}}{Z} = \langle V \rangle \quad (54)$$

$$\frac{\partial^2 \ln Z}{\partial (\beta p)^2} = -\frac{1}{Z^2} \left(\frac{\partial Z}{\partial \beta p} \right)^2 + \frac{1}{Z} \frac{\partial^2 Z}{\partial (\beta p)^2} = -\langle V \rangle^2 + \langle V^2 \rangle = \langle (\delta V)^2 \rangle \quad (55)$$

$$= -\frac{\partial \langle V \rangle}{\partial \beta p} = -k_B T \left(\frac{\partial \langle V \rangle}{\partial p} \right)_{N,T} \quad (56)$$

$$\frac{\langle (\delta V)^2 \rangle}{\langle V \rangle} = -k_B T \frac{1}{V} \left(\frac{\partial \langle V \rangle}{\partial p} \right)_{N,T} = k_B T \kappa_T \quad (57)$$

where κ_T is the isothermal compressibility.

2.4 The grand canonical ensemble

In the thermodynamic limit ($N \rightarrow \infty$), the microcanonical ensemble becomes comparable to the canonical ensemble but only considers mechanical work rather than chemical work. We assume that a system can remain at a constant chemical potential μ by interacting with a reservoir R at a stable temperature. The probability of all reservoir states is determined by adding all possible microstates.

$$P(\nu) = \begin{cases} \frac{e^{-\beta(E-\mu N)}}{\Xi} & \text{if } T_\nu = T, V_\nu = V, \mu_\nu = \mu \\ 0 & \text{otherwise} \end{cases} \quad (58)$$

where the grand partition function is summation by grouping all microstates with a given number of particles.

$$\Xi(T, \mu, V) = \sum_\nu e^{-\beta(E-\mu N)} = \sum_{N=0}^{\infty} e^{\beta \mu N} \sum_{(\mu|N)} e^{-\beta E} \quad (59)$$

The unconditional probability of finding N particles in the grand canonical system can be reduced by the results from canonical ensemble.

$$P(N) = \frac{e^{\beta\mu N} Q(N, V, T)}{\Xi(T, \mu, V)} \quad (60)$$

The sharpness of the distribution for N results that the sum in grand partition function can be approximately by its largest term at $N = N^* = \langle N \rangle$. Then the grand potential is defined on the basis of the T, μ, V ensemble.

$$\Xi(T, \mu, V) = \lim_{N \rightarrow \infty} \sum_{N=0}^{\infty} e^{\beta\mu N} Q(N, V, T) = e^{\beta\mu N^*} Q(N^*, V, T) = e^{\beta\mu N - \beta F} = e^{-\beta \Gamma} \quad (61)$$

$$\Gamma(T, \mu, V) = E - TS - \mu N = -pV = -k_B T \ln \Xi(T, \mu, V) \quad (62)$$

The average and fluctuation of energy can be calculated from the grand partition function, similar to the equation 46.

$$\frac{\partial \Xi}{\partial \beta \mu} = \sum_{\nu} N e^{-\beta(E - \mu N)} \quad \text{and} \quad \frac{\partial^2 \Xi}{\partial (\beta \mu)^2} = \sum_{\nu} N^2 e^{-\beta(E - \mu N)} \quad (63)$$

The cumulants of N are generated by $\ln \partial \Xi(\beta \mu)$.

$$\langle N \rangle = \frac{\partial \ln \Xi}{\partial \beta \mu} = \frac{\partial \beta p V}{\partial \beta \mu} = V \left(\frac{\partial p}{\partial \mu} \right)_{T, V} \quad (64)$$

$$\langle (\delta N)^2 \rangle = \frac{\partial^2 \ln \Xi}{\partial (\beta \mu)^2} = \frac{\partial \langle N \rangle}{\partial \beta \mu} = k_B T \left(\frac{\partial N}{\partial \mu} \right)_{T, V} \quad (65)$$

$$\frac{\langle (\delta N)^2 \rangle}{\langle N \rangle} = \frac{k_B T}{V} \left(\frac{\partial N}{\partial p} \right)_{T, V} = -k_B T \left(\frac{\partial N}{\partial V} \right)_{p, T} \frac{1}{V} \left(\frac{\partial V}{\partial p} \right)_{N, T} = n k_B T \kappa_T \quad (66)$$

where n is the number density of the system and κ_T is the isothermal compressibility.